

NICOLE SARNA

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EDUCATION

Doctor of Philosophy (PhD), Bioengineering

Rice University, Houston, TX
Vanderbilt University, Nashville TN

Expected: Summer 2026

Jul. 2024-Present

Aug. 2021-Jun. 2024

Dissertation: Leveraging T Cell Mechanotransduction and Membrane Dynamics to Enhance Activation and Function

Selected Honors/Awards: NSF Graduate Research Fellowship Program (GRFP)

Master of Science (MS), Biomedical Engineering

Vanderbilt University, Nashville, TN

Aug. 2021-May 2024

Thesis: Quantitative Analysis of Circulating Tumor Cell Dynamics in Patients Undergoing Pluvicto Therapy for Metastatic Prostate Cancer

Bachelor of Science (BS), Biomedical Engineering

University of Florida, Gainesville, FL

Aug. 2017-May 2021

Selected Honors/Awards: Magna Cum Laude, Outstanding Undergraduate Research Award, Bright Futures Scholarship

PROFESSIONAL EXPERIENCES

Graduate Research Fellow

Rice University

Aug. 2021-Present

Houston, TX

Advisor: Michael R. King, PhD, Department of Bioengineering

- Innovating ex vivo CAR T cell manufacturing strategies that leverage mechanical stimuli to enhance activation, persistence, and cytotoxicity at solid tumor sites following adoptive T cell transfer therapy
- Isolating primary human T cells and conducting advanced biophysical assays to reveal age- and sex-specific differences in membrane composition and mechanical properties to inform personalized immunotherapy strategies
- Developing a computational model of T cell activation (ODE-based intracellular signaling) to generate predictive insights into how membrane composition and biophysical forces regulate immune cell signaling and function

Graduate Research Fellow (Visiting Scholar)

Vanderbilt University

Aug. 2021-Jun. 2024

Nashville, TN

Advisor: Michael R. King, PhD, Department of Biomedical Engineering

- Led a translational research project investigating circulating tumor cell (CTC) dynamics in metastatic prostate cancer (mPCa) patients undergoing Pluvicto (^{177}Lu -PSMA-617) therapy, collaborating closely with oncologists, radiologists, and faculty at Vanderbilt University Medical Center to align experimental design and data interpretation with clinical context
- Processed over 100 blood samples from mPCa patients before, during, and after Pluvicto treatment to isolate and immunostain CTCs, enabling temporal analysis of CTC burden and phenotypic changes throughout treatment
- Developed an image processing algorithm to quantify PSMA protein expression and nuclear features in CTCs to correlate cellular characteristics with clinical markers of treatment response, including PSA levels and radiographic findings

Undergraduate Research Fellow

University of Florida

Jul. 2019-Aug. 2021

Gainesville, FL

Advisor: Carlos M. Rinaldi-Ramos, PhD, Department of Chemical and Biomedical Engineering

- Characterized super-paramagnetic iron oxide nanoparticles (SPIONs) for *in vivo* imaging applications in cancer immunotherapy
- Evaluated the sensitivity and resolution of in-house synthesized SPIONs using Magnetic Particle Imaging (MPI) system
- Performed *in vivo* experiments to monitor and track the biodistribution of T cells following Adoptive Cell Transfer (ACT) in breast cancer and glioblastoma murine models

R&D Project Lead

Lucere Laboratories, LLC

Sept. 2020-Mar. 2021

Gainesville, FL

Supervisor: Atticus Steinmetz, Founder & CEO

- Led the development of a luminometer prototype to optimize detection and reduce manufacturing costs of D-Luciferin
- Established critical partnerships, including with the UF Brain Tumor Immunotherapy Dept, enabling product testing and optimization
- Developed and executed a marketing and sales strategy leveraging CRM tools to identify target markets, tailor products to client needs, and drive collaboration between academic and industrial sectors

Undergraduate Research Fellow

University of Florida

Jan. 2020-May 2020

Gainesville, FL

Advisor: Todd E. Golde, MD, PhD, Department of Neuroscience

- Developed MATLAB program to analyze fluorescent images of 3D ex vivo brain slice cultures that exhibit aggregation of tau protein, a primary marker of Alzheimer's and other neurodegenerative diseases

Undergraduate Research Fellow

University of Florida

Sept. 2017-Dec. 2018

Gainesville, FL

Advisor: Norman Fitz-Coy, PhD, Department of Mechanical and Aerospace Engineering

- Collected data for a collaborative research project, DebrisSat, led by NASA, The Aerospace Corporation, and the US Air Force Space and Missile Systems Center to improve and update NASA's Standard Breakup Model through orbital debris modeling
- Analyzed space debris from hypervelocity satellite collisions to inform spacecraft design and improve debris mitigation strategies

TECHNICAL SKILLS

Research Techniques:

- Flow cytometry, cell sorting (FACS)
- Light, fluorescence, confocal, TIRF, IRM microscopy
- Micropipette aspiration and optical tweezers
- Immunoblot/Western blot, SDS-PAGE/gel electrophoresis
- Animal handling and in vivo experiments
- Rotary microtome, immunohistochemistry (IHC/IF)
- IVIS SpectrumCT, Magnetic Particle Imaging (MPI)
- Dynamic Light Scattering (DLS)
- Dynamic Magnetic Susceptibility (DMS)

Human Subject Research:

- Human blood collection, sample processing, and cell culture (isolation of immune cells and circulating tumor cells)
- Phlebotomy (Texas School of Phlebotomy)
- Lentiviral transduction and non-viral transfection
- Institutional Review Board (IRB) and grant proposal writing

Data Analysis & Computational Tools:

- FlowJo, ImageJ, GraphPad Prism, Excel
- MATLAB, COMSOL, CSS/HTML, Git, Python
- Solidworks, Onshape, Autodesk Fusion

PATENTS

Fluid Shear Stress for Ex Vivo Activation Of Immune Effector Cells

Issued: Oct. 3, 2024

Patent Application No: US 2024/0327792 A1

Inventors: King, MR, Hope, JM, Dombroski, JA, Sarna, NS.

PUBLICATIONS

* Asterisk = Co-First Author

- Desai, S.H.* , Sarna N.S.*, Luo, A.C., King, M.R. “**Fluoxetine Acts as a Calcium-Dependent Sensitizer of TRAIL-Induced Apoptosis in Glioblastoma Multiforme**”, *ACS Omega* (2026). doi: 10.1109/RBME.2024.3505073
- Sarna, N.S., Curry, N.M., Aalei, E., Kaufman, B.G., King, M.R. “**Immunomechanobiology: Engineering the Activation and Function of Immune Cells with the Mechanical Signal of Fluid Shear Stress**”, *IEEE Reviews in Biomedical Engineering* (2025). doi: 10.1109/RBME.2024.3505073
- King S.M.*, Sarna N.S.*, Ortiz I.*, Wang W., Lopez-Cavestany M., Zhang Z. “**Retention of E-Selectin Functionalized Liposome Fanny Packs on Jurkat Cells Following Invasion through Collagen**”, *Journal of Immunological Methods* (2024). doi: 10.1016/j.jim.2024.113700
- Sarna, N.S., Desai, S.H., Kaufman, B.G., Hanna, A.M., King, M.R. (2024). “**Enhanced and Sustained T cell Activation in Response to Fluid Shear Stress**” *iScience* (2024). doi: 10.1016/j.isci.2024.109999
- Knoblauch, S.V., Desai, S.H., Dombroski, J.A., Sarna, N.S., Hope, J.M., King, M.R. “**Chemical and Mechanical Activation of Piezo1 Enhance TRAIL-Mediated Apoptosis in Glioblastoma Cells**”, *ACS Omega*, (2023). doi: 10.1021/acsomega.3c00705
- Sarna, N.S.*, Marrero-Morales, L*, DeGross, R*, Rivera-Rodriguez, A, Lui, S, Chiu-Lam, A, Good, H, Rinaldi-Ramos, C.M. “**An anatomically correct 3D printed mouse phantom for magnetic particle imaging studies**”, *Bioengineering & Translational Medicine* (2022). doi: 10.1002/btm2.10299
- Dombroski, J.A.*, Hope, J.M.*, Sarna, N.S., King, M.R. “**Channeling the Force: Piezo1 mechanotransduction in cancer metastasis**”, *Cells* (2021). doi: 10.3390/cells10112815
- Rivera-Rodriguez, A, Hoang-Minh, L, Chiu-Lam, A, Sarna, N.S., Marrero-Morales, L, Mitchell, D, Rinaldi-Ramos, C.M. “**Tracking Adoptive T Cell Immunotherapy Using Magnetic Particle Imaging**” *Nanotheranostics* (2021). doi: 10.7150/ntno.55165
- Lui, S*, Rivera-Rodriguez, A*, Chiu-Lam, A*, DeGross, R, Savliwala, S, Sarna, N.S., Rinaldi-Ramos, C.M. “**Long Circulating Tracer Tailored for Magnetic Particle Imaging**” *Nanotheranostics* (2021). doi: 10.7150/ntno.58548

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TEACHING AND MENTORSHIP EXPERIENCE

Graduate Student Research Mentor

Jan. 2022-Present

Rice and Vanderbilt University

Houston, TX | Nashville, TN

- Mentor and train undergraduate and high school students on research techniques in Dr. Michael King's lab
- Conceptualize research project which aims to treat chemotherapy resistant glioblastoma brain cancer cells through combined treatment regimens and improved immune cell delivery
- Design, plan, and oversee experiments performed by mentees

SyBBURE Searle Undergraduate Research Program

Jan. 2022-Jun. 2024

Vanderbilt University

Nashville, TN

- SyBBURE Searle Graduate Fellow
 - Lead weekly subgroup meetings with undergraduate students to provide guidance and direction in their research topics
 - Advise and mentor a group of undergraduate students through a semester long, team-based STEM project
 - Organize and teach personal and professional skill workshops for undergraduate students, including Computer Aided Design (CAD), computer programming, circuit board design, CV/resume building, and time and stress management

Biomedical Engineering Lab (BME2900/3900/4901) Teaching Assistant

Jan. 2022-May 2022

Vanderbilt University

Nashville, TN

- Aided sophomore (3 sections), junior (1 section), and senior (1 section) undergraduate BME students with experimental design, data collection in lab, and scientific writing
- Provided detailed feedback, edits, and grades to student drafts and final lab reports

Introduction to Engineering (ES1041) Teaching Assistant

Aug. 2021-Dec. 2021

Vanderbilt University

Nashville, TN

- Assisted freshman undergraduate students with their coursework and final projects that involve BME wearable device design conceptualization and prototyping
- Prepared slide decks and led lectures on microcontroller programming and Computer Aided Design (CAD) for project prototyping

CONFERENCES & PRESENTATIONS

- **[Oral] Sarna, NS**, Hanna, AH, Desai, SH, Antunovic, M, Schaffer, KR, Hurley, PJ & King, MR. "Circulating Tumor Cell Dynamics in Patients Undergoing Pluvicto Therapy for Metastatic Prostate Cancer", **Biomedical Engineering Society (BMES) Annual Meeting**, Baltimore, MD (Oct. 2024)
- **[Oral] Sarna, NS** & King, MR. "Modeling T Cell Calcium Dynamics and Downstream Activation Signaling in Response to Fluid Shear Stress", **Biomedical Engineering Society (BMES) Annual Meeting**, Baltimore, MD (Oct. 2024)
- **[Oral] Sarna, NS**, Desai, SH, Kaufman, BK, Curry, NM, Hanna, AM & King, MR. "Enhanced and Sustained T Cell Activation, Proliferation, and Cytotoxicity in Response to Fluid Shear Stress", **Biomedical Engineering Society (BMES) Annual Meeting**, Baltimore, MD (Oct. 2024)
- **[Oral] Sarna, NS**, Desai, SH, & King, MR. "Leveraging T Cell Mechanotransduction for Enhanced and Sustained Ex Vivo Activation", **Biomedical Engineering Society (BMES) Annual Meeting**, Seattle, WA (Oct. 2023)
- **[Poster] Sarna, NS**, Desai, S, & King, MR. "Leveraging T cell mechanotransduction for enhanced and sustained ex vivo activation", **NIH National Cancer Institute (NCI) Tissue Engineering Collaborative (TEC) Annual Meeting**, Portland, OR (Aug. 2023)
- **[Poster] Sarna, NS**, Desai, S, & King, MR. "Leveraging T cell mechanotransduction for enhanced and sustained ex vivo activation", **The Southeastern Immunology Symposium**, Nashville, TN (Jun. 2023)
- **[Poster] Sarna, NS**, Hope, JM, Desai, S, & King, MR. "Enhanced T Cell Activation via Fluid Shear Stress", **The Vanderbilt Institute of Nanoscale Science and Engineering (VINSE) Annual Meeting**, Nashville, TN (Nov. 2022)
- **[Oral] Sarna, NS**, Hope, JM, Desai, SH, & King, MR. "Enhanced and Sustained T Cell Activation Over Time via Fluid Shear Stress Exposure", **Biomedical Engineering Society (BMES) Annual Meeting**, San Antonio, TX (Oct. 2022)
- **[Poster] Sarna, NS**, Marrero-Morales, L, DeGroff, R, Rivera-Rodriguez, A, Lui, S, Chiu-Lam, A, Good, H, Rinaldi-Ramos, CM. "An Anatomically Correct 3D-Printed Mouse Phantom for Magnetic Particle Imaging Studies", **World Molecular Imaging Congress (WMIC) Annual Meeting**, Miami, FL (Sept. 2022)
- **[Poster] Sarna, NS**, Hope, JM, Desai, S, & King, MR. "Enhanced T Cell Activation via Fluid Shear Stress", **The Vanderbilt Center on Mechanobiology Inaugural Retreat**, Nashville, TN (Aug. 2022)
 - 2nd place Poster Presentation

- **[Poster] Sarna, NS**, Hope, JM, Desai, S, & King, MR. "Enhanced T Cell Activation via Fluid Shear Stress", **NIH National Cancer Institute (NCI) Tissue Engineering Collaborative (TEC) Annual Meeting**, *Madison WI* (July 2022)
 - 1st place Poster Presentation
- **[Oral] Sarna, NS**, Marrero-Morales, L, DeGroff, R, Rivera-Rodriguez, A, Lui, S, Good, H, Rinaldi-Ramos, CM. "Advancing the Principles of Replacement, Reduction, and Refinement by Evaluating an Anatomically Correct Mouse Phantom for a Brain Tumor Model in Magnetic Particle Imaging", **University of Florida Undergraduate Research Symposium**, *Gainesville, FL* (Apr. 2021)
 - Outstanding Undergraduate Research Award
- **[Poster] Sarna, NS**, Marrero-Morales, L, R, Rivera-Rodriguez, Rinaldi-Ramos, CM. "Evaluating the Sensitivity of the Momentum™ Magnetic Particle Imaging System for Ferucarbotran Iron Oxide Nanoparticles" **American Institute of Chemical Engineers (AIChE) Annual Meeting**, *Orlando, FL* (Nov. 2019)

UNIVERSITY INVOLVEMENT

Vanderbilt University

Nashville, TN

- BME Graduate Student Association (GSA)
 - Member (Aug. 2021-Present)
 - Co-chair, Elementary Education Outreach (Aug. 2021-Dec. 2021)
 - Chair, Elementary Education Outreach (Jan. 2022-Aug. 2023)

University of Florida

Gainesville, FL

- Biomedical Engineering Society, Member (Aug. 2017-May 2021)
- Society of Women in Engineering (SWE), Member (Aug. 2017-May 2021)
- University of Florida Philharmonic Orchestra, Violinist (Aug. 2017-Dec. 2017)

PROFESSIONAL DEVELOPMENT CONFERENCES

- **[Attendee] NextProf Pathfinder Workshop**, *San Diego, CA* (Oct. 2022)
- **[Attendee] IEEE Engineering in Medicine and Biology Conference (EMBC)**, *Orlando, FL* (Aug. 2016)